

Electric Field Distribution in Black Silicon Needles Under Reverse Bias: A TCAD Simulation Study

Oskari Leiviskä^{1*}, Olli Setälä¹, Ville Vähänissi¹, and Hele Savin¹

¹Department of Electronics and Nanoengineering, Aalto University

*oskari.leiviska@aalto.fi

Introduction

- Nanotextured silicon surface (black silicon, b-Si) has been used to reduce optical loss < 1 % in silicon-based photodetectors.
- Electric field distribution in black silicon morphologies mostly unexplored under high operation voltages.
- This work provides more insight on how b-Si affects on electric fields within reverse biased devices.

Methods

- Simulated simplified b-Si needle array with Silvaco ATLAS TCAD. Basic needle geometry obtained from¹.
- No additional dopants introduced to b-Si; the surface layer inverted to hole majority due to negative charge near the Al₂O₃/Si-interface.

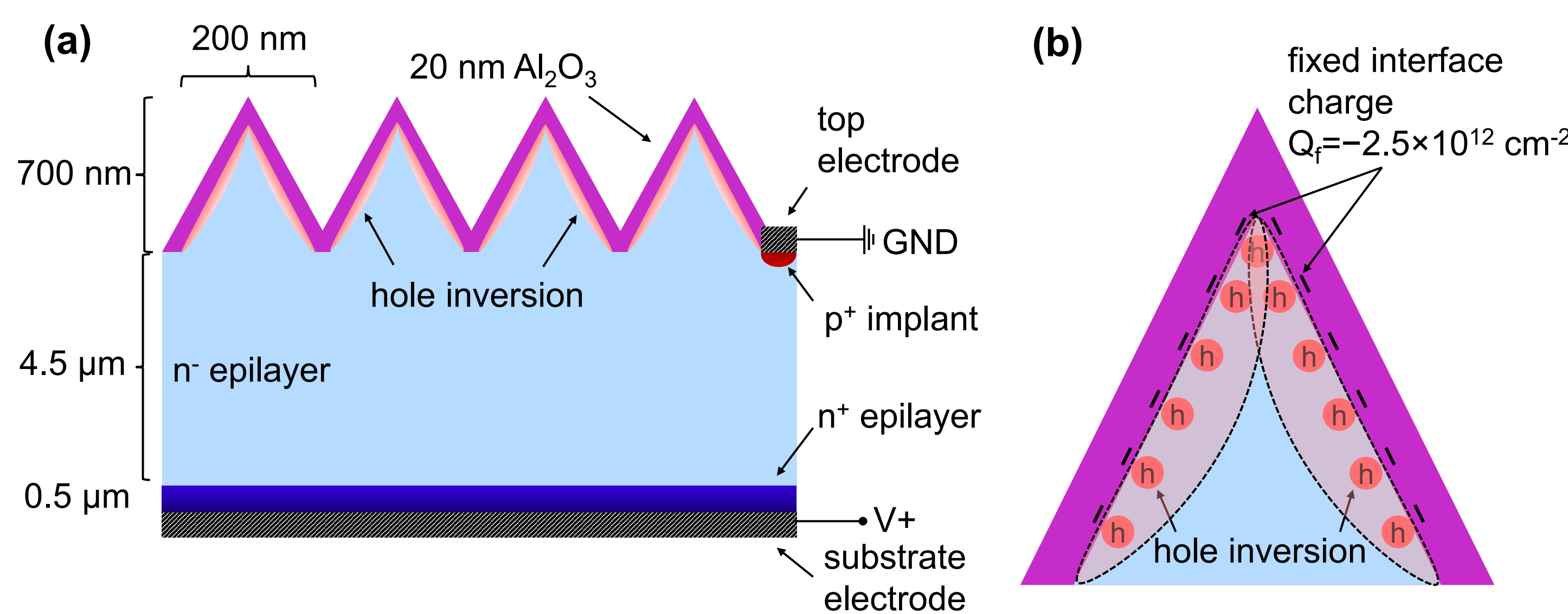


Figure 1. (a) Cross section of used TCAD structure. (b) cross-section of black silicon needle showing the hole inversion caused by fixed charge present near Al₂O₃/Si-interface.

Electric field and impact generation under high bias

- Inside the b-Si needles, the electric field remains almost constant as bias is increased from 0 V to 105 V.
- Impact ionization is limited to bulk of n⁺ epilayer; charge carriers are not multiplied by avalanche effects inside b-Si needle array.

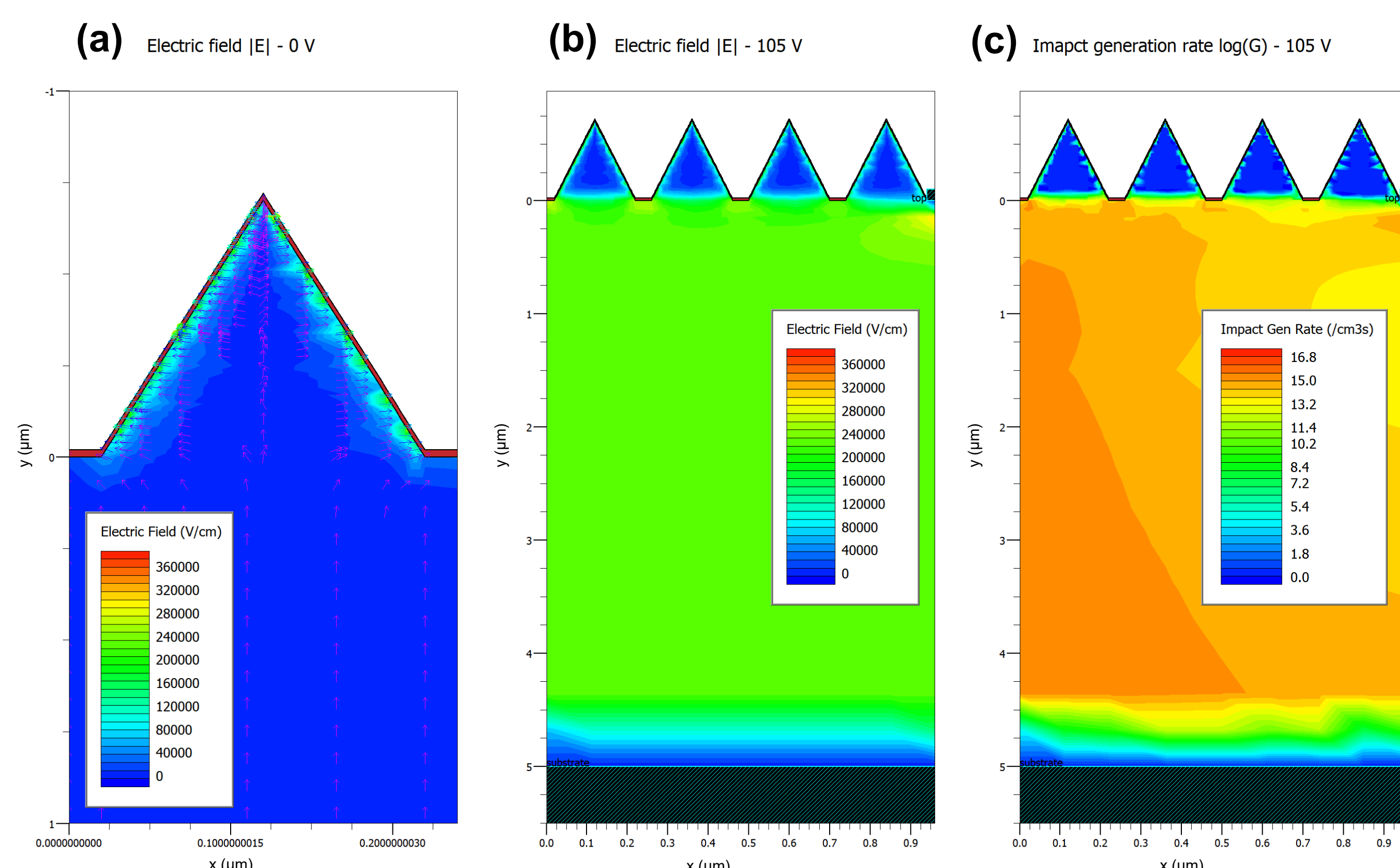


Figure 2. (a) Electric field magnitude and direction inside and below a b-Si needle at 0 V bias. Electric field extends uniformly to y=5 μm (not shown here). (b) Electric field in the structure at 105 V bias. The geometry prevents field from penetrating inside the b-Si needles. (c) Impact generation profile (carrier avalanching) in the structure at 105 V bias. The charge carriers are not multiplied inside the b-Si needles.

Impact of needle geometry

- Needle width and height was varied to see the impacts of geometry on electric field profile inside b-Si needles.
- Wider base width resulted in better field penetration inside the bulk of b-Si needles.

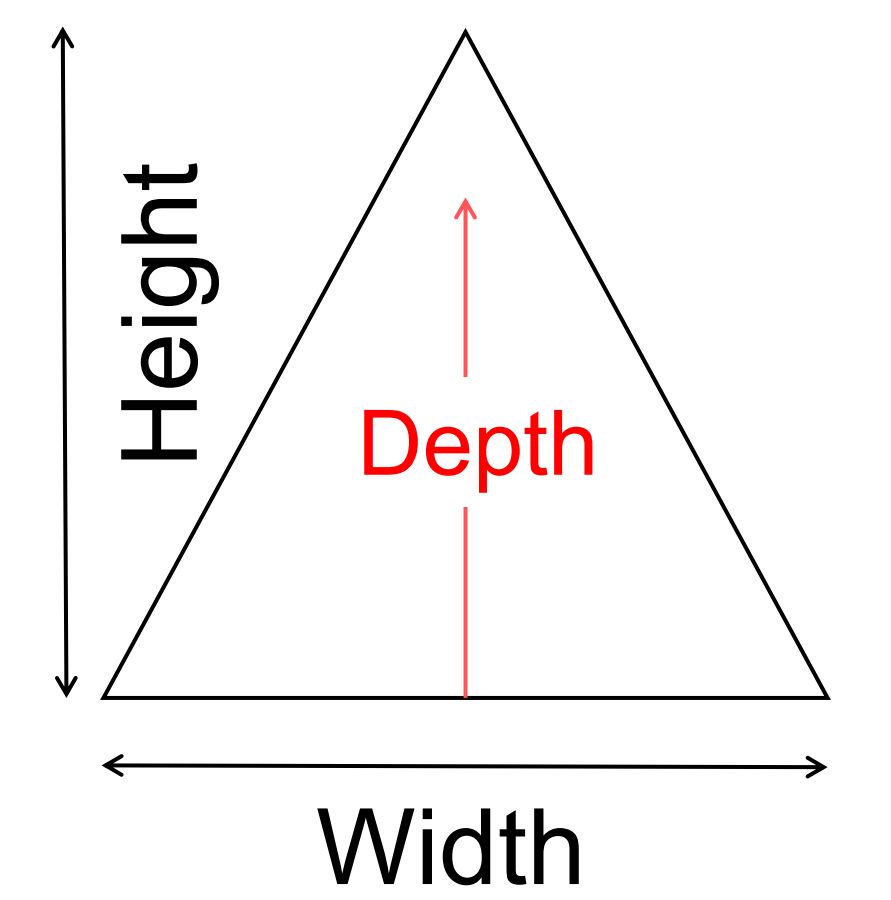


Figure 2. Illustration of needle dimensions. Depth is measured from the base towards tip of the needle and used as x-axis in Figure 3.

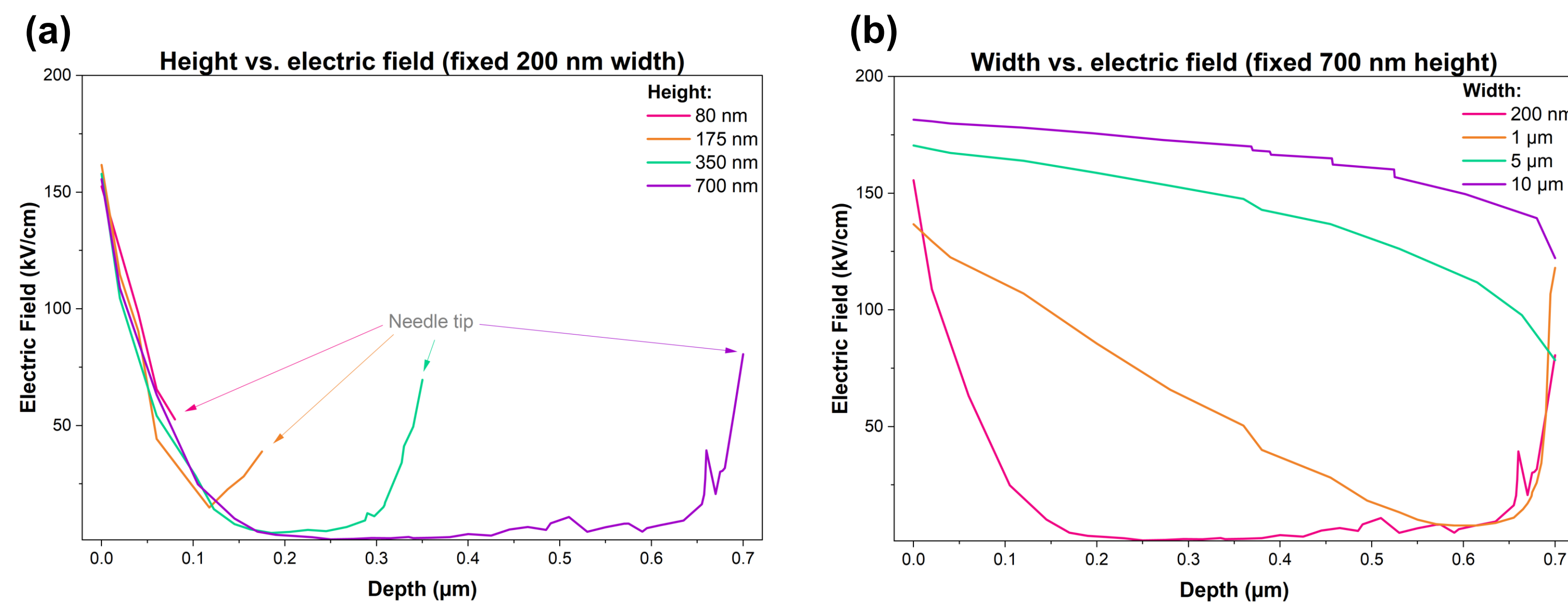


Figure 3. Total electric field [E] inside the needle measured at increasing depth (from bottom towards tip) at 105 V reverse bias. (a) With varying needle heights at fixed base width of 200 nm. (b) With varying needle base widths at fixed height of 700 nm.

Impact on gain in b-Si avalanche photodiodes

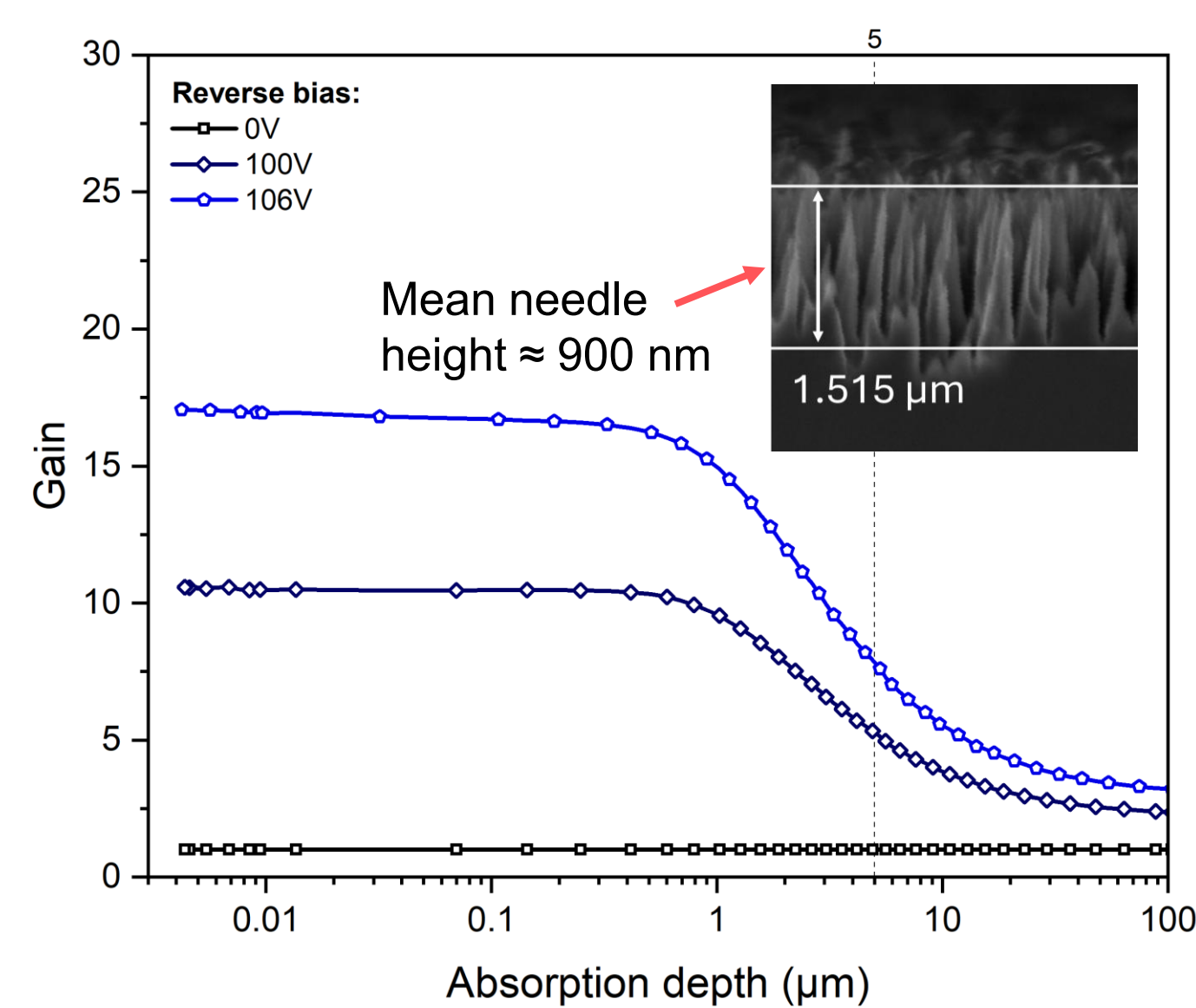


Figure 4. Measured illumination wavelength dependent avalanche gain using different reverse bias voltages. Wavelengths are converted to corresponding optical absorption depths³. Note: ~5 μm device thickness is show with dashed gray line. Inset: SEM image of b-Si on top of the avalanche photodiode.

- Experimental avalanche photodiodes with b-Si positioned on surface of uniform multiplication region².
- Gain should increase towards shallower absorption depths (increased electron path length in avalanche region)
- Gain saturates for short wavelengths which are absorbed inside the b-Si needles ($\delta < 1 \mu\text{m}$) → b-Si acts as absorber/drift region without impact generation

Summary

- Impact of black silicon to electric field profiles was studied under high voltage TCAD simulations
- Electric field caused by external voltage does not penetrate inside b-Si needles before needle geometry becomes drastically flatter.

References

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